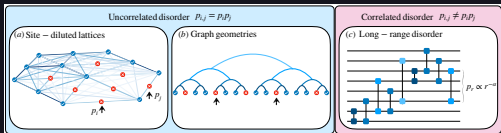


Scalable spin squeezing on complex quantum networks

Why scalable spin squeezing matters

- Quantum sensing demands **robust** entanglement
- Real devices suffer from **disorder & noise**
- Understanding **network effects** is key to scalability

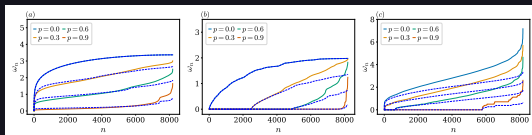
Real-world impact of disorder



Disorder types in experiments:

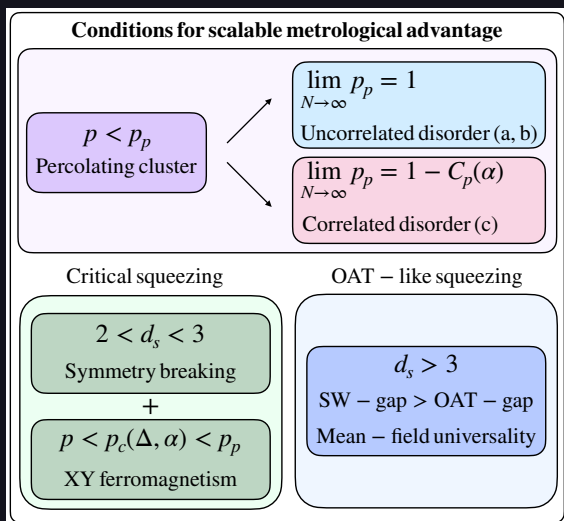
- Site dilution in lattices (ions, Rydberg, molecules)
- Engineered graph connectivity (programmable arrays)
- Spatially correlated errors in quantum processors

The key idea: XXZ model on graphs



- Spin waves governed by graph Laplacian spectrum
- Collective zero mode acts like one-axis twisting rotor

Spin squeezing mechanisms



Two distinct routes to **scalable squeezing** governed by **spectral dimension** d_s :

- **OAT-like squeezing** for $d_s > 3$: mean-field rotor dominates dynamics
- **Critical squeezing** for $2 < d_s < 3$: spontaneous symmetry breaking enabled by transient random walks

Impressive results: scaling regimes

Phase sensitivity scaling
beyond shot noise:
 $(\delta\phi)^2 \sim N^{-(1+\mu)}$

- OAT:

$$\mu = 2/3,$$

$$\nu = 1/3$$

scaling,

long-

range

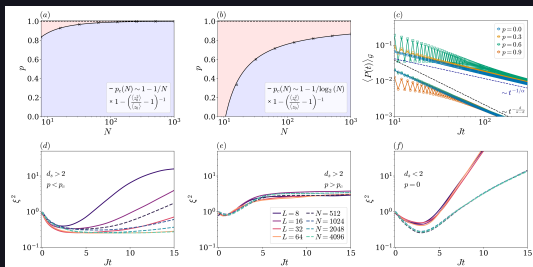
con-

nected

graphs

- Critical:

Towards real quantum sensors



- **Trapped ions** — tunable exponent & dilution
- **Rydberg arrays** — engineered graphs and cavity control
- **Solid-state defects** — disorder and dipolar interaction

Takeaway

Scalable spin squeezing
depends universally on
spectral dimension and
disorder structure.

Designing quantum networks with
spectral dimension $d_s > 2$ and strong
connectivity
is essential for robust quantum metrology.

Towards scalable quantum sensors on noisy,
complex platforms.